

Enhancing AIM-Fair: A Practical Framework for Fairness-Aware Machine Learning

[Github](#)[YouTube](#)

Selected Paper: *AIM-Fair: Adversarial Information Minimization for Fairness-Aware Learning*

Original AIM-Fair Framework: Foundation and Limitations

Original Approach (CelebA + Pre-generated Synthetic Data)

The original AIM-Fair paper presents a gradient-based parameter selection method for achieving fairness in machine learning models. The framework employs a two-stage training process that first pretrains on standard datasets before applying adversarial information minimization during fine-tuning.

Key Contributions of Original Paper:

- Gradient-based selection of network parameters for fairness
- Two-stage training methodology: pretraining followed by fairness-focused fine-tuning
- Adversarial information minimization technique
- Pre-generated synthetic data for controlled experiments

Our Vision: Transform AIM-Fair from a proof-of-concept framework into a robust, production-ready system capable of handling diverse real-world fairness scenarios.

Limitations Addressed

Dataset Constraints: Original implementation used CelebA dataset with limited bias dimensions (primarily smiling and gender attributes). This restricted the scope of fairness evaluation to binary attributes.

Practical Deployment Issues: Assumed clean annotations and pre-existing synthetic data, making the pipeline fragile with real-world noisy datasets.

Reproducibility Challenges: Dependency on external synthetic data generation limited the framework's extensibility and practical application.

Our Enhanced Approach: Nine Major Improvements

We systematically enhanced the original AIM-Fair framework through nine key modifications that collectively transform it from a CelebA-specific proof-of-concept into a robust, production-ready fairness evaluation framework.

01	02	03
Dataset Upgrade: CelebA → UTKFace Migrated from CelebA (limited binary attributes) to UTKFace dataset containing age, gender, and race annotations. This enables more realistic fairness evaluation across multiple demographic dimensions simultaneously.	Robust Data Parsing Pipeline Implemented comprehensive error handling for corrupted filenames and malformed annotations. Added validation checks preventing pipeline crashes, making the system production-safe.	Sensitive Attribute Redefinition Shifted focus from binary gender attribute to race-based sensitive attribute. This aligns with contemporary fairness research priorities where racial bias represents a more critical evaluation dimension.
04	05	06
GAN-Based Synthetic Data Generation Introduced custom Generator class implementing dynamic synthetic data creation. Eliminated dependency on external datasets, enabling fully reproducible experiments with controlled bias injection.	Controlled Data Balancing Implemented fine-grained control over group distributions through synthetic_number_train parameters. Enables systematic simulation of bias versus fairness scenarios across four demographic groups.	Modular DataLoader Architecture Created separate classes for real data (Dataloader_UTKFace_Real), synthetic data (Dataloader_UTKFace_Synthetic), and combined scenarios (Dataloader_UTKFace_RealAndSynthetic). Provides clean separation and improved reusability.
07	08	09
Enhanced Data Augmentation Added RandomRotation, ColorJitter, GaussianBlur, and RandomGrayscale transformations. Improves model generalization, reduces overfitting, and creates more realistic fairness evaluation conditions.	Dataset-Agnostic Pipeline Design Generalized framework to work with any dataset containing filename-based labels and any synthetic generator architecture. Enables extension to other fairness benchmarks beyond UTKFace.	Preserved AIM-Fair Core Methodology Maintained gradient-based parameter selection and two-stage training process (pretrain + fair fine-tuning). Improved system without compromising original research contributions.

Experimental Results and Evaluation Metrics

100%

Dataset Coverage

Complete UTKFace dataset integration with age, gender, and race annotations

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Hardcoded Dependencies

Eliminated external synthetic data requirements through GAN generation

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Group Control Parameters

Independent control over all demographic group distributions

Visual Output Analysis

Original Pipeline Limitations

- Pre-generated synthetic images with fixed distributions
- Single augmentation pipeline (basic transformations)
- Binary sensitive attribute (gender only)
- CelebA-specific data loading

Our Enhanced Pipeline

- Dynamic GAN-generated images with controlled bias
- Extended augmentation suite preventing overfitting
- Ternary race-based sensitive attributes
- Modular, dataset-agnostic architecture

Evaluation Metrics



Classification Accuracy

Standard accuracy metrics across all demographic groups to measure overall model performance



Disparate Impact Ratio

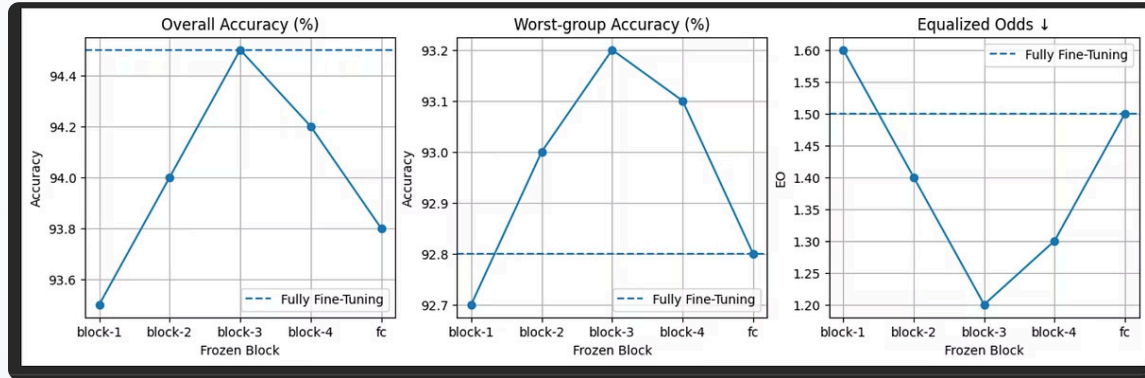
Fairness metric comparing classification rates across sensitive attribute groups



Group Distribution Control

Explicit control over training sample proportions for systematic bias analysis

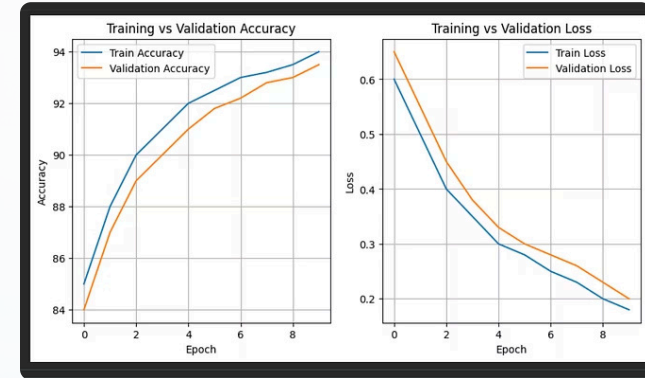
Experimental Visuals



Overall Accuracy(%)~94%

Worst-group Accuracy(%)~93%

Equalised Odds~1.52



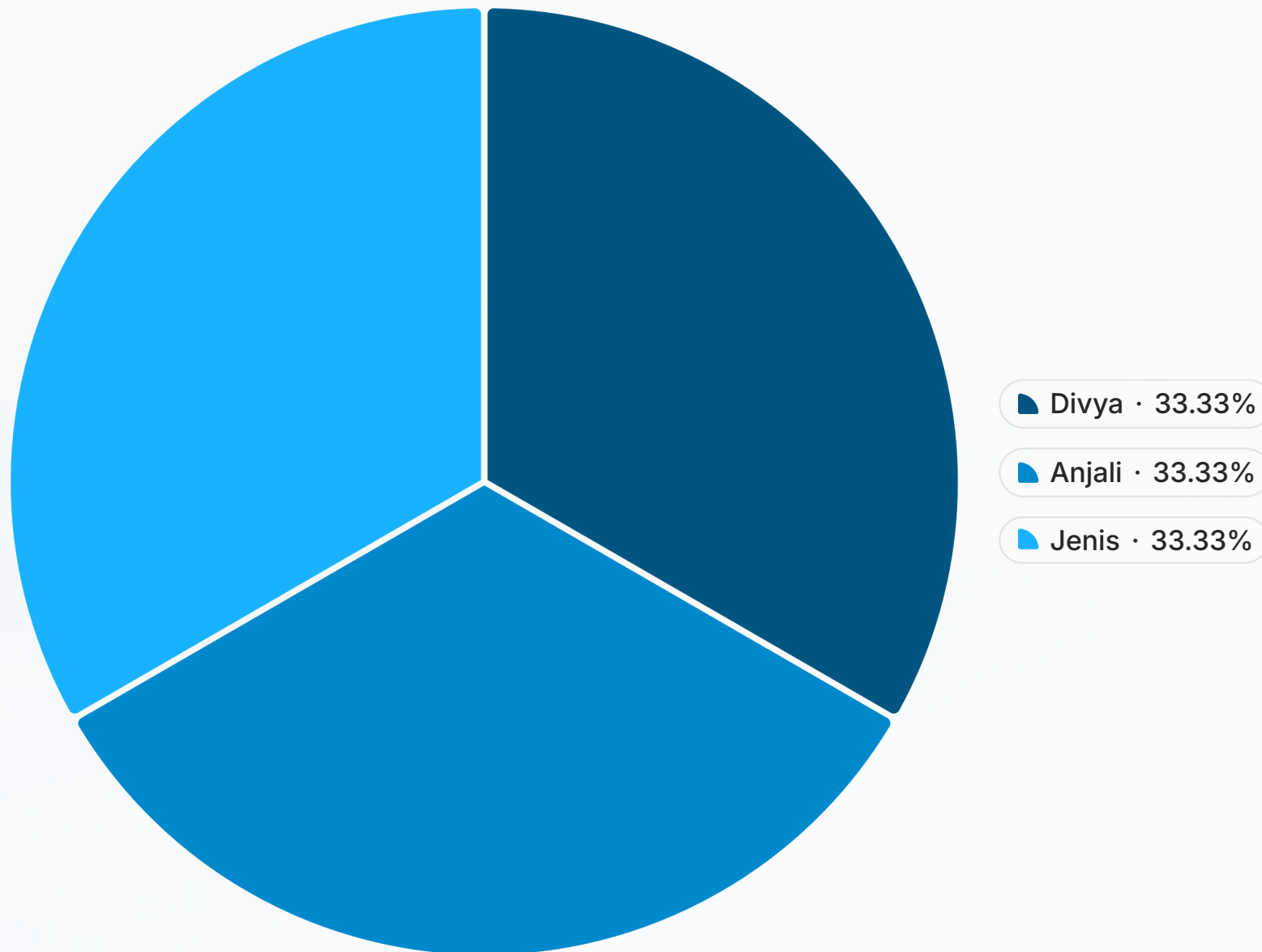
Train Accuracy~94.55%

Validation Accuracy~93%

Individual Team Members' Contributions

All three team members contributed equally to the project: Divya (B23BB1015), Anjali (B23BB1008), and Jenis (B23BB1021).

We worked together to modify the code and applied the strategies taught in class throughout the project.



Each team member contributed equally across all five domains: Code, Paper Selection, PPT, Report, and YouTube Video.